Pre-Calculus 11 - Trigonometry



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

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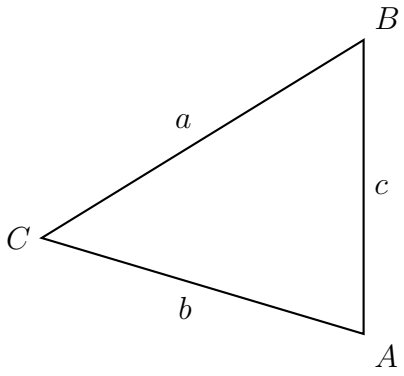
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1 Triangle Area and Sine Law

When working with non-right triangles, we can still rely on our standard right triangle trigonometry (SOHCAHTOA), and some general algebra skills.

Common Notation

In a triangle, we usually label our vertices using capital letters. Then the side opposite to the vertex is usually denoted by the same letter, but lower case.



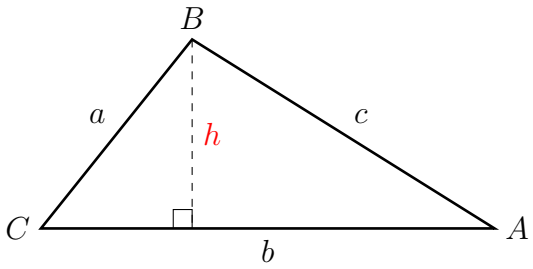
Angles can be identified as $\angle A = \angle BAC = \angle CAB$.
Sides can also be referenced by their endpoints, hence

$$AC = b, \quad BC = a, \quad AB = c$$

Lastly, common variables for angles include

- θ: Theta
- β: Beta
- ϕ: Phi
- α: Alpha
- γ: Gamma
- ψ : Psi

Example 1.1. Show that the area of the given triangle is $Area = \frac{1}{2}ab\sin C$.



The SAS Area Formula

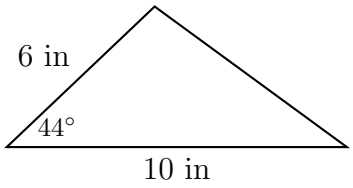
In any triangle, where two sides and the ***included angle*** is known, the area can be calculated by the formula

$$Area = \frac{1}{2}ab\sin C$$

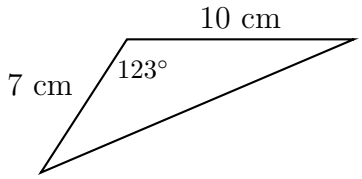
Where a, b are the two sides and $\angle C$ is the included angle.

Example 1.2. Find the area of the given triangles. Round to the nearest hundredth.

a)



b)



Example 1.3. Using the SAS Area formula, show that

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Hint: $Area = \frac{1}{2}ab\sin C \quad Area = \frac{1}{2}ac\sin B \quad Area = \frac{1}{2}bc\sin A$

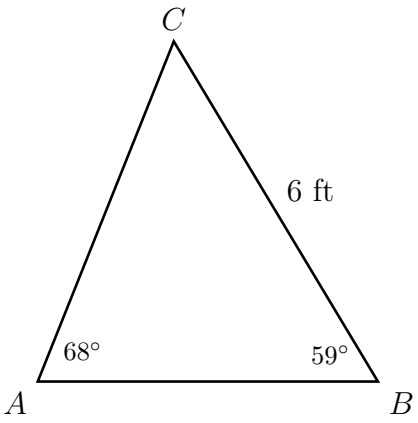
The Law of Sines

In triangle $\triangle ABC$,

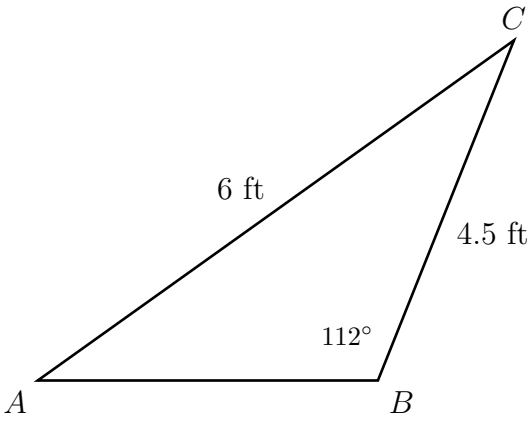
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

This simply means that the ratio of the sine of an angle and its opposite side, is proportional to any other angle and its opposite side. Hence, the reciprocal version holds true too.
In short, you need to have a "known pair" (angle + opposite side), and then some other part of the triangle (side or angle). Then you can set up a proportion equation to solve.

Example 1.4. Find the length of AC.



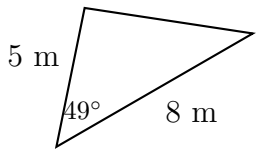
Example 1.5. Find $\angle A$.



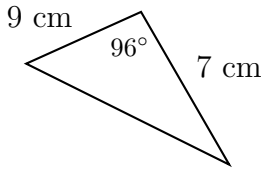
Practice Exercises

1. Find the area of each triangle. Round to the nearest hundredth.

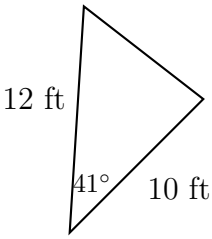
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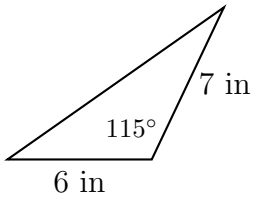
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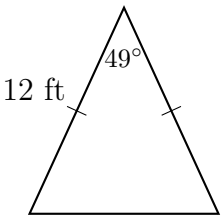
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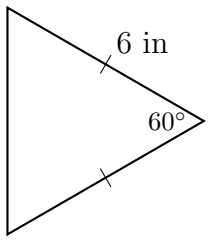
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e)

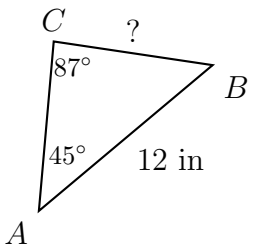


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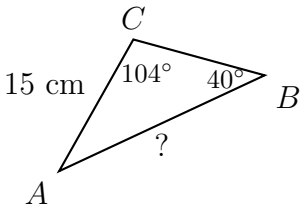


2. Find the missing side length in each triangle.

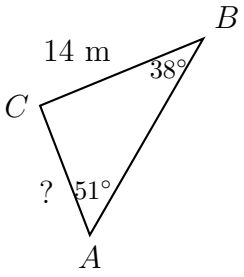
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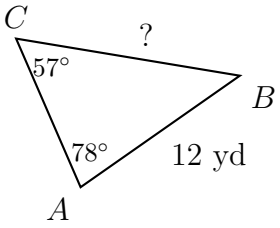
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c)

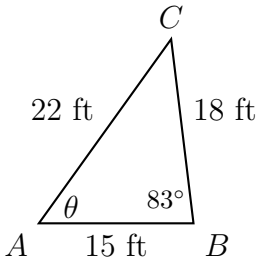


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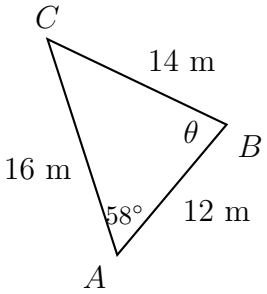


3. Find the measure of the indicated angle θ in each triangle.

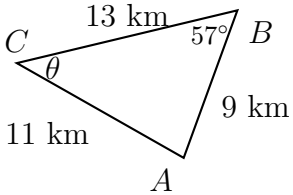
a)



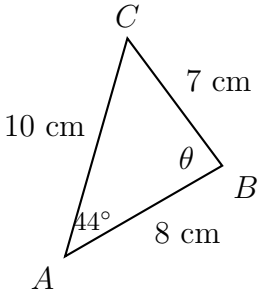
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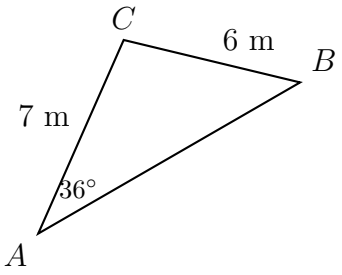
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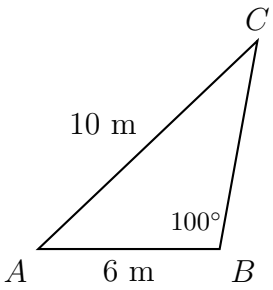
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4. Solve the triangle (find all missing sides and angles).



5. Solve the triangle (find all missing sides and angles).



2 Cosine Law and Applications

In triangles where you know two sides and the included angle (SAS), or where you know all three sides (SSS), you can use the **Cosine Law** to help solve the triangle.

The Law of Cosines

In triangle $\triangle ABC$,

$$c^2 = a^2 + b^2 - 2ab\cos C$$

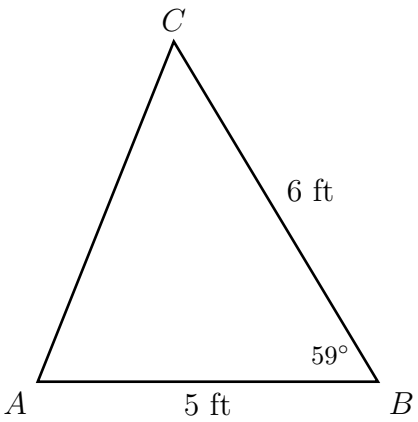
Similarly,

$$a^2 = b^2 + c^2 - 2bc\cos A \quad \text{and} \quad b^2 = a^2 + c^2 - 2ac\cos B$$

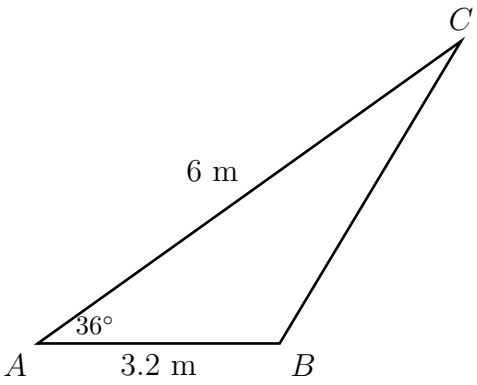
The Cosine Law is especially useful when you *do not* have a "known pair" (as in the Sine Law), but instead know either:

- Two sides and the included angle (SAS) to find the third side, or
- All three sides (SSS) to find an angle.

Example 2.1. Find the length of AC.



Example 2.2. Find $\angle A$.



Example 2.3. Show that if $a^2 = b^2 + c^2 - 2bc\cos A$, then $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

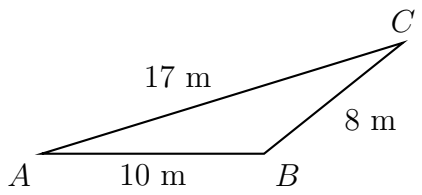
Cosine Law for Angles

If we know all three sides of a triangle, we can use the following to find any angle

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

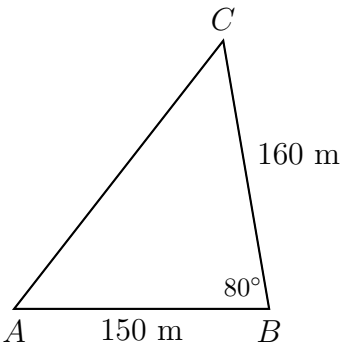
In general, it is recommended that if you start with a cosine law application, that you stick to solving the triangle with cosine law. There is sometimes an ambiguous case with sine law, that is harder to identify. We will identify this case later.

Example 2.4. Find all the angles in the triangle below. Round to the nearest tenths.



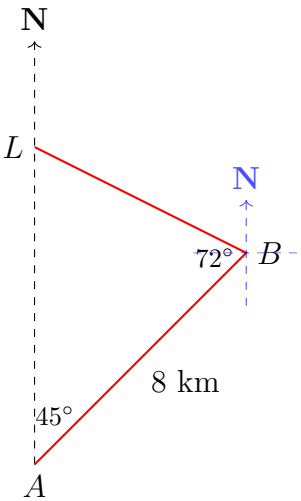
Application Problems

Example 2.5. A surveyor measures a triangular plot of land ABC. Side $AB = 150$ m, side $BC = 160$ m, and $\angle B = 80°$. Find the length of side AC and the area of the plot.



Example 2.6. In travelling a flat stretch of the highway, you observe a mountain directly in front of you. The angle of elevation to the top of the mountain is 4.3°. After driving for 25 km towards the mountain, the angle of elevation to the top changes to 6.7°. How high is the mountain?

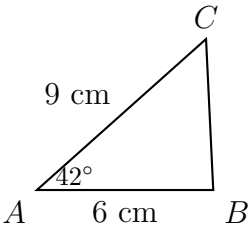
Example 2.7. A ship at point A spots a lighthouse L due north. The ship travels 8 km on a bearing of 045° to point B. From B, the lighthouse now appears on a bearing of 297°. How far is the ship from the lighthouse at point B?



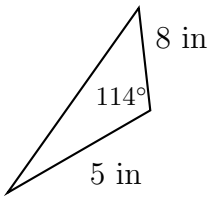
Practice Exercises

1. Find the missing side length in each triangle.

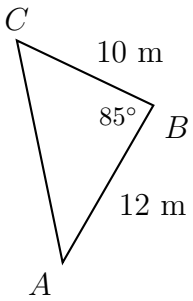
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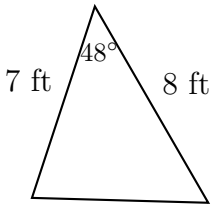
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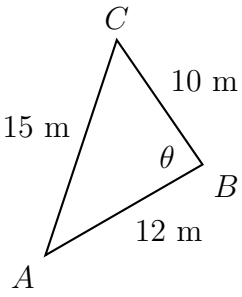


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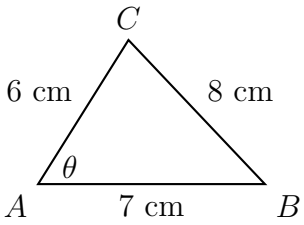


2. Find the measure of the indicated angle θ in each triangle.

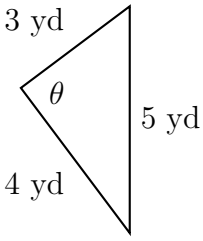
a)



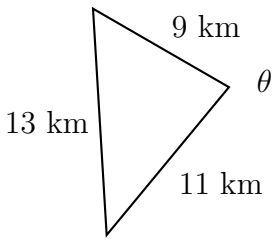
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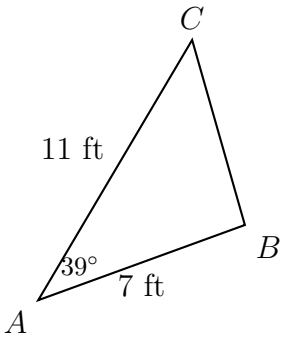
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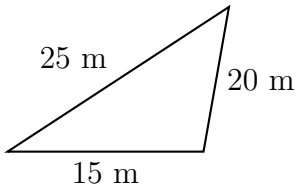
d)



3. Solve the triangle (find all missing sides and angles).



4. A triangular garden has sides of length 15 m, 20 m, and 27 m. Find the measure of the largest angle in the garden.



5. Two ships leave port at the same time. Ship A travels at 15 knots on a bearing of 065°, while Ship B travels at 12 knots on a bearing of 155°. How far apart are the ships after 2 hours?

6. A baseball diamond is a square with sides 90 feet long. The pitcher's mound is 60.5 feet from home plate. How far is the pitcher's mound from first base?

7. A triangular parcel of land has sides measuring 120 m, 150 m, and 180 m. A fence needs to be built along the longest side. What angle should the fence make with the 120 m side at their meeting point?

8. An aircraft flies 200 km on a bearing of 120°, then turns and flies 150 km on a bearing of 210°. How far is the aircraft from its starting point?

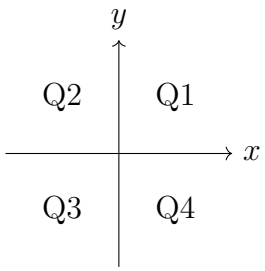
3 Angles in Standard Position

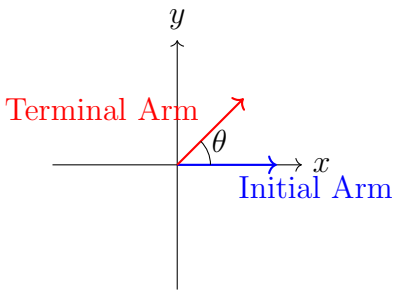
When we expand our view of angles beyond just triangles, we need to have some common understanding.

Angles in Standard Position

An angle is in **standard position** when:

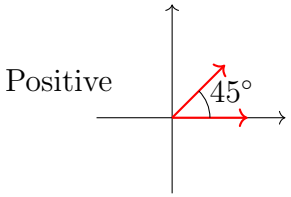
- Its vertex is at the origin $(0, 0)$
- Its **initial arm** is **fixed** and lies along the positive x-axis
- Its **terminal arm** rotates from the initial arm

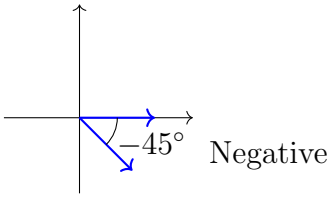




Direction of Rotation

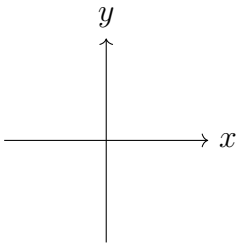
- **Positive angles**: Counterclockwise rotation
- **Negative angles**: Clockwise rotation



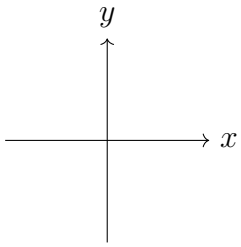


Example 3.1. Sketch the following angles in standard position.

a) $190°$



b) $-300°$



Coterminal Angles

Angles that share the same terminal arm are called **coterminal**.

For example, $60°$, $-300°$, and $420°$ are all coterminal angles.

To find coterminal angles: **Add or subtract** $360°$

$$\theta_{\text{coterminal}} = \theta + 360°n \quad \text{(where } n \text{ is any integer)}$$

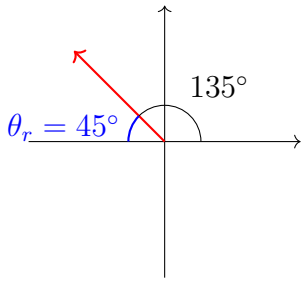
Example 3.2. For each angle below, give one positive and one negative coterminal angle.

a) $200°$

b) $-100°$

Reference Angles

The **reference angle** is the acute angle between the terminal arm and the x-axis. They are used to help us find the ratios for any angle.



- Always positive and acute ($0° < \theta_{\text{ref}} < 90°$)

Example 3.3. Find the reference angle to the following angles in standard position.

a) $210°$

b) $-60°$

c) $160°$

d) $400°$

Example 3.4. Find all angles $0° < \theta < 360°$, such that the reference angle is $67°$.

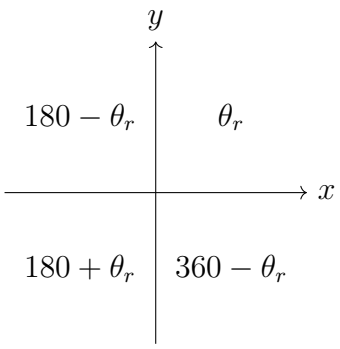
Principal Angle and Reference Angle Shortcuts

We define a **principal angle** as a coterminal angle that lies between $0°$ and $360°$. This allows us to work with angles we're much more familiar with and can visualize easier, while still retaining all the same properties.

Quick idea to find reference angle

1. Is the angle closest to $0°, 180°,$ or $360°$?
2. How close?

Once you know the reference angle, you can find the standard angles in the other quadrants by



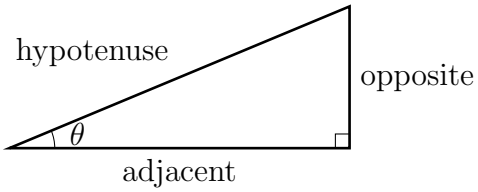
Example 3.5. Find the reference angle of the given θ in standard position. Then find the three other principal angles with the same reference angle.

a) $910°$

b) $-400°$

Trigonometric Ratios on a Circle

Recall: The trigonometric ratios on a right triangle



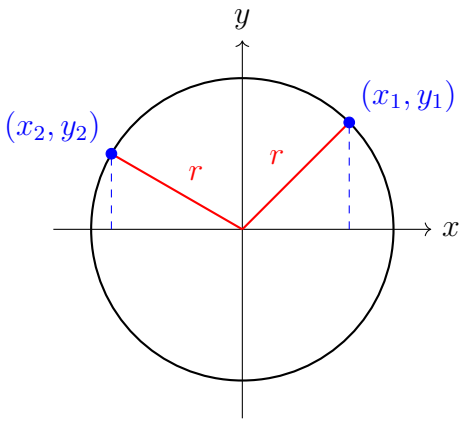
$$\sin\theta = \frac{opp}{hyp} \qquad \cos\theta = \frac{adj}{hyp} \qquad \tan\theta = \frac{opp}{adj}$$

The Equation of a Circle

The general equation of a circle on a coordinate plane is

$$x^2 + y^2 = r^2$$

Where r is the radius and the centre of the circle is at $(0, 0)$.



As the terminal arm moves along the circle, it can form a right triangle with the x-axis. We then use the reference angle to determine our trigonometric ratios. So another way to think about the trigonometric ratios is

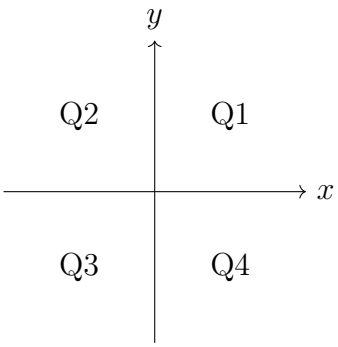
$$\sin\theta = \frac{y}{r} \qquad \cos\theta = \frac{x}{r} \qquad \tan\theta = \frac{y}{x}$$

This allows us to visualize how the ratios work beyond just acute angles.

Example 3.6. The point $P(-3,-4)$ lies on the terminal arm of an angle θ in standard position. Determine the exact values of $\sin\theta, \cos\theta$, and $\tan\theta$.

Notice how some trigonometric ratios are positive and some are negative...

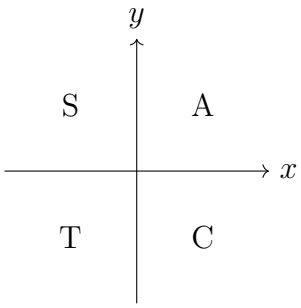
Fill out the table below by determining if the values should be **positive or negative** in the respective quadrants



Quadrant	x	y	$\sin\theta$	$\cos\theta$	$\tan\theta$
I					
II					
III					
IV					

The ASTC Rule

- Quadrant I, **A**ll the trig ratios are positive.
- Quadrant II, only **S**ine is positive.
- Quadrant III, only **T**angent is positive.
- Quadrant IV, only **C**osine is positive.



A simple mnemonic device to help you remember is "**A**ll **S**tudents **T**ake **C**alculus".

Example 3.7. Determine which quadrant(s) an angle θ in standard position would terminate in given

a) $\cos\theta < 0$ b) $\tan\theta > 0, \sin\theta < 0$ c) $\sin\theta > 0, \cos\theta < 0$

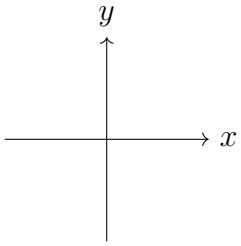
Example 3.8. The point $P(5, -\sqrt{42})$ lies on the terminal arm of an angle θ in standard position. Determine the exact values of $\sin\theta, \cos\theta$, and $\tan\theta$.

Example 3.9. Given $\sin\theta = -\frac{4}{7}$ and $\cos\theta < 0$ for an angle θ in standard position. Determine the exact values of $\cos\theta$ and $\tan\theta$.

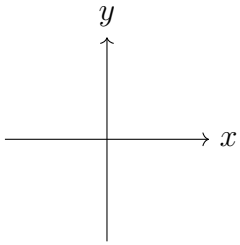
Practice Exercises

1. Sketch the following angles in standard position.

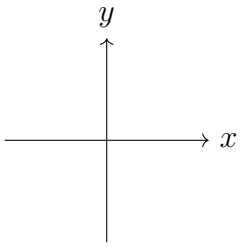
a) $135°$



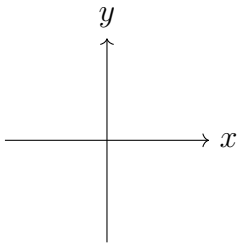
b) $-210°$



c) $300°$



d) $-45°$



2. For each angle below, give one positive and one negative coterminal angle.

a) $45°$

b) $-150°$

c) $315°$

d) $-75°$

3. Find the reference angle for each angle in standard position.

a) $150°$

b) $-120°$

c) $225°$

d) $-30°$

e) $330°$

f) $-210°$

g) $100°$

h) $500°$

4. Find all angles $0° < \theta < 360°$ with the given reference angle.

   a) Reference angle $= 30°$

   b) Reference angle $= 55°$

   c) Reference angle $= 15°$

   d) Reference angle $= 72°$

5. Find the reference angle and three other principal angles with the same reference angle.

   a) $500°$

   b) $-150°$

   c) $780°$

   d) $-600°$

6. The following points lie on the terminal arm of an angle θ in standard position. Determine the exact values of $\sin\theta$, $\cos\theta$, and $\tan\theta$.

   a) $P(6, -8)$

   b) $P(-5, 12)$

   c) $P(-1, -1)$

   d) $P(2\sqrt{3}, 2)$

7. Determine which quadrant(s) an angle θ in standard position would terminate in given the conditions.

   a) $\sin\theta > 0$ and $\tan\theta < 0$

   b) $\cos\theta > 0$ and $\tan\theta > 0$

   c) $\sin\theta < 0$ and $\cos\theta < 0$

   d) $\tan\theta < 0$ and $\cos\theta > 0$

   e) $\sin\theta < 0$ and $\tan\theta > 0$

   f) $\cos\theta < 0$ and $\tan\theta < 0$

8. Given the trigonometric ratios, determine the exact values of the other ratios.

   a) $\sin\theta = \frac{3}{5}$ and $\cos\theta < 0$

   b) $\cos\theta = -\frac{12}{13}$ and $\tan\theta > 0$

   c) $\tan\theta = \frac{4}{3}$ and $\sin\theta < 0$

   d) $\sin\theta = -\frac{5}{13}$ and θ in Quadrant III

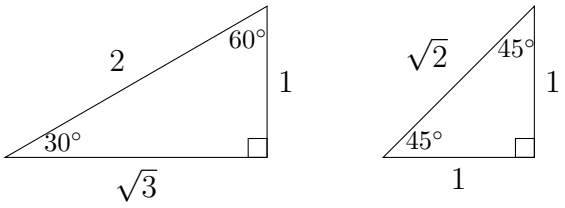
   e) $\tan\theta = -\frac{7}{6}$ and $\sin\theta < 0$

   f) $\cos\theta = \frac{6}{7}$ and θ in Quadrant IV

4 Special Angles

In trigonometry, we have a few special angles that we commonly use because they can give us an exact value. They stem from the 30-60-90 and the 45-45-90 triangles.

Special Triangles



Example 4.1. Complete the table below for the given trigonometric functions and angles.

	30°	45°	60°
$\sin\theta$			
$\cos\theta$			
$\tan\theta$			

Example 4.2. Draw the angle and then evaluate. State what you notice.

a) $\sin 150°$ b) $\sin 210°$ c) $\sin 330°$

Example 4.3. Draw the angle and then evaluate. State what you notice.

a) $\cos 135°$ b) $\cos 225°$ c) $\cos 315°$

Finding Exact Value Quickly

These steps can help you find an exact value quickly, without a calculator.

1. Determine the quadrant the standard angle is in.
2. Determine the reference angle of the given angle.
3. Use the ASTC rule to determine if the given function is positive or negative.
4. The ratio is identical to the ratio given by the reference angle.

Example 4.4. Find the exact value of the following. Without a calculator.

a) $\tan 120°$

b) $\sin 225°$

c) $\cos(-60°)$

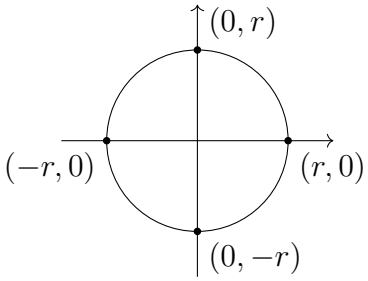
d) $\cos 210°$

e) $\sin 390°$

f) $\tan 150°$

Quadrantal Angles

Angles that land exactly on the axes: $0°, 90°, 180°, 270°, 360°$



No reference angles - use the x, y, r ratios directly!

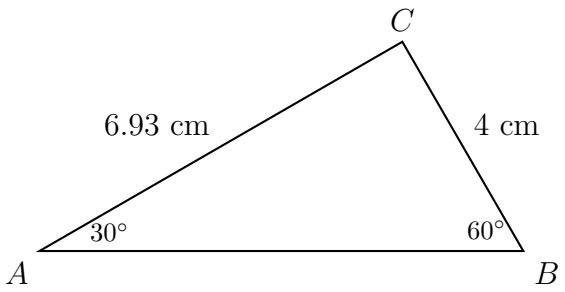
Example 4.5. Find the exact value of the following. Without a calculator.

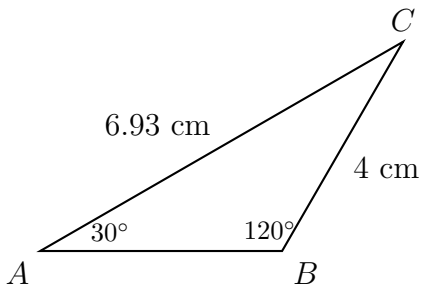
a) $\tan 90°$ b) $\cos 180°$ c) $\sin 270°$ d) $\tan 0°$

The Ambiguous Case of the Sine Law

Recall that in a triangle, if you have a known pair and another side, you can use Sine Law to solve the triangle.

Example 4.6. For each triangle below, state the given values of $\angle A, a$, and b.



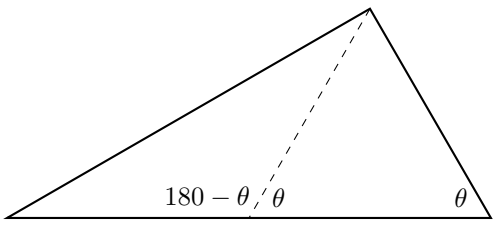


We can see if without a drawing, just stating the properties of a triangle that appears to be solvable by Sine Law, may come across as **ambiguous**.

This stems from the property that we have seen in the ASTC Rule,

$$\sin\theta = \sin(180 - \theta)$$

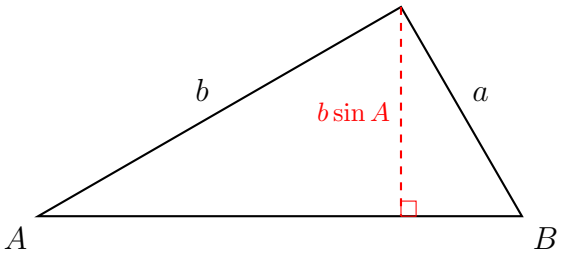
Recall that in both Q1 and Q2, sine is positive.



Ambiguous Sine Law Inequality

If given a known pair ($\angle A$ and a) and some other side, b, on the triangle. Then an ambiguous case can exist if and only if

$$b\sin A < a < b$$



If $a < b\sin A$, then no such triangle is possible.

Example 4.7. Using the inequality property, determine the number of possible triangles.

a) $\angle A = 40°$, $a = 10$ cm, $b = 12$ cm

b) $\angle C = 70°$, $c = 10$ cm, $b = 12$ cm

c) $\angle X = 100°$, $y = 10$ cm, $x = 12$ cm

Example 4.8. Solve for triangle(s) ABC if $\angle A = 30°$, $a = 6$ cm, and $b = 7$ cm.

Practice Exercises

1. Draw the angles and then evaluate exactly.

a) $\sin 120°$ b) $\sin 240°$ c) $\sin 300°$

d) $\cos 150°$ e) $\cos 210°$ f) $\cos 330°$

g) $\tan 135°$ h) $\tan 225°$ i) $\tan 315°$

2. Find the exact value of the following without a calculator.

a) $\sin 135°$ b) $\cos 150°$ c) $\tan 210°$ d) $\sin 330°$

e) $\cos 225°$ f) $\tan 300°$ g) $\sin(-45°)$ h) $\cos(-120°)$

i) $\tan 495°$ j) $\sin 750°$ k) $\cos 510°$ l) $\tan(-150°)$

3. Find the exact value of the quadrantal angles without a calculator.

a) $\sin 90°$ b) $\cos 180°$ c) $\tan 270°$ d) $\sin 0°$

e) $\cos 270°$ f) $\tan 180°$ g) $\sin 180°$ h) $\cos 90°$

i) $\tan 0°$ j) $\sin 270°$ k) $\cos 360°$ l) $\tan 90°$

4. Using the ambiguous case inequality property, determine the number of possible triangles.

a) $\angle A = 35°$, $a = 8$ cm, $b = 12$ cm b) $\angle B = 50°$, $b = 15$ cm, $c = 20$ cm

c) $\angle C = 25°$, $c = 5$ cm, $a = 10$ cm d) $\angle A = 110°$, $a = 18$ cm, $b = 12$ cm

e) $\angle B = 40°$, $b = 7$ cm, $a = 5$ cm f) $\angle C = 60°$, $c = 8$ cm, $b = 9$ cm

g) $\angle A = 80°$, $a = 6$ cm, $b = 4$ cm h) $\angle B = 30°$, $b = 12$ cm, $c = 24$ cm

5. Solve for all possible triangles ABC given the following information.

a) $\angle A = 40°$, $a = 8$ cm, $b = 10$ cm

b) $\angle B = 35°$, $b = 12$ cm, $c = 15$ cm

c) $\angle A = 25°$, $a = 5$ cm, $b = 8$ cm

d) $\angle C = 50°$, $c = 10$ cm, $b = 12$ cm

5 Solving Trigonometric Equations

We now have the full understanding to solve trigonometric equations of multiple forms.

Example 5.1. Given $\sin 30° = \frac{1}{2}$, what value(s) of θ will have $\sin\theta = \frac{1}{2}$?

The answer to this may surprise you...

Solving θ between $0° \leq \theta < 360°$

One of the fundamental skills we will need to develop is to solve for the primary angles, θ, that satisfies the trigonometric equation.

1. Using ASTC rule, determine which quadrants the angle must be in.
2. Based on the ratio, determine the reference angle.
3. Find the equivalent standard angle with said reference angle, in the respective quadrants.

Example 5.2. Solve for θ, where $0° \leq \theta < 360°$. Try not to use a calculator.

a) $\cos\theta = \frac{1}{2}$

b) $\tan\theta = -\sqrt{3}$

c) $\sin\theta = -\frac{\sqrt{2}}{2}$

d) $\cos\theta = -\frac{\sqrt{3}}{2}$

Solving for Non-Special Angles (with calculator)

Not all ratios will be solvable without a calculator. The steps are similar, except we need a calculator to find the reference angle.

1. Using ASTC rule, determine which quadrants the angle must be in.
2. Take the inverse function of the **positive ratio** to get the reference angle. Round as needed.
3. Find the equivalent standard angle with said reference angle, in the respective quadrants.

Example 5.3. Solve for θ, where $0° \leq \theta < 360°$. Round θ to the nearest tenths.

a) $\cos\theta = 0.67$

b) $\tan\theta = -6.7$

Solving for all angles, θ

Every solvable trigonometric equation has infinite solutions over an all real domain.

1. Find the two different quadrant angles that solves the trigonometric equation.
2. Add or subtract 360° to find other answers.

Answers can be written in the form

$$\theta = \theta_1 + 360n, \theta_2 + 360n$$

Where θ_1, θ_2 are the two different quadrant solutions, and $n \in \mathbb{Z}$.

Note: There are some nuances to this method, as it is the ***period*** of the function that matters. For now, we will assume the **period** is 360°.

Interesting fact: The **period** of a tangent function is actually 180°.

Example 5.4. Find the general solution to the following.

a) $\sin\theta = \frac{1}{2}$

b) $\tan\theta = -\sqrt{3}$

Example 5.5. Solve for θ, where $-180° \leq \theta < 180°$.

a) $\sin\theta = -\frac{\sqrt{3}}{2}$

b) $\tan\theta = -\frac{\sqrt{3}}{3}$

Example 5.6. Solve for θ, where $-90° \leq \theta \leq 90°$.

a) $\cos\theta = -\frac{1}{\sqrt{2}}$

b) $\sin\theta = \frac{\sqrt{3}}{2}$

Quadrantal Angles

Angles that are on the axes don't quite fit the mould we've built.

1. Determine which angle(s) $0°, 90°, 180°, 270°$ satisfy the equation. Drawing the circle helps.
2. Adjust angles as needed to satisfy the given domain.

Example 5.7. Solve for θ, where $0° \leq \theta < 360°$. Try not to use a calculator.

a) $\cos\theta = 1$

b) $\tan\theta = 0$

c) $\sin\theta = 0$

d) $\tan\theta$ is undefined.

Advanced Forms

Once you've mastered the basic forms, we can move into more advanced forms that may require additional steps before solving.

Example 5.8. Solve for θ, where $0° \leq \theta < 360°$.

a) $2\cos\theta - 1 = 0$

b) $\sqrt{3}\tan\theta + 1 = 0$

Example 5.9. Solve for θ, where $0° \leq \theta < 360°$.

a) $\tan\theta(\cos\theta + 1) = 0$

b) $2\sin^2\theta + \sin\theta - 1 = 0$

note : $\sin^2\theta = (\sin\theta)^2$

Practice Exercises

1. Solve for θ, where $0° \leq \theta < 360°$. Find exact values without a calculator.

   a) $\sin\theta = \frac{\sqrt{3}}{2}$ b) $\cos\theta = \frac{1}{2}$ c) $\tan\theta = 1$

   d) $\sin\theta = -\frac{1}{2}$ e) $\cos\theta = -\frac{\sqrt{2}}{2}$ f) $\tan\theta = -\sqrt{3}$

   g) $\sin\theta = \frac{\sqrt{2}}{2}$ h) $\cos\theta = -\frac{1}{2}$ i) $\tan\theta = \frac{\sqrt{3}}{3}$

   j) $\sin\theta = -\frac{\sqrt{3}}{2}$ k) $\cos\theta = \frac{\sqrt{3}}{2}$ l) $\tan\theta = -1$

2. Solve for θ, where $0° \leq \theta < 360°$. Round to the nearest tenth of a degree.

   a) $\sin\theta = 0.8$ b) $\cos\theta = -0.45$ c) $\tan\theta = 2.5$

d) $\sin\theta = -0.33$ e) $\cos\theta = 0.92$ f) $\tan\theta = -0.75$

g) $\sin\theta = 0.15$ h) $\cos\theta = -0.68$ i) $\tan\theta = 5.2$

3. Find the general solution for each equation (all angles θ).

a) $\sin\theta = \frac{1}{2}$ b) $\cos\theta = -\frac{\sqrt{3}}{2}$

c) $\tan\theta = 1$ d) $\sin\theta = -\frac{\sqrt{2}}{2}$

e) $\cos\theta = \frac{1}{\sqrt{2}}$ f) $\tan\theta = -\sqrt{3}$

g) $\sin\theta = 0$ h) $\cos\theta = -1$

4. Solve for θ in the specified domain.

a) $\sin\theta = -\frac{1}{2}$, $-180° \leq \theta < 180°$

b) $\cos\theta = \frac{\sqrt{3}}{2}$, $-180° \leq \theta < 180°$

c) $\tan\theta = -1$, $-180° \leq \theta < 180°$

d) $\sin\theta = \frac{\sqrt{3}}{2}$, $-90° \leq \theta \leq 90°$

e) $\cos\theta = -\frac{1}{2}$, $-90° \leq \theta \leq 90°$

f) $\tan\theta = \sqrt{3}$, $-180° \leq \theta < 180°$

5. Solve for quadrantal angles, where $0° \leq \theta < 360°$.

a) $\cos\theta = 1$

b) $\sin\theta = 0$

c) $\tan\theta = 0$

d) $\sin\theta = 1$

e) $\cos\theta = -1$

f) $\tan\theta$ is undefined

6. Solve the advanced forms for θ, where $0° \leq \theta < 360°$.

a) $2\sin\theta - 1 = 0$

b) $\sqrt{3}\cos\theta + 1 = 0$

c) $2\tan\theta - 2 = 0$

d) $4\cos\theta + 2 = 0$

e) $3\sin\theta + 3 = 0$

f) $\sqrt{2}\cos\theta - 1 = 0$

g) $2\tan\theta + 2\sqrt{3} = 0$

h) $4\sin\theta - 2\sqrt{3} = 0$

7. Solve by factoring for θ, where $0° \leq \theta < 360°$.

a) $\sin\theta(\cos\theta - 1) = 0$

b) $\tan\theta(\sin\theta + 1) = 0$

c) $\cos\theta(2\sin\theta - 1) = 0$

d) $(2\cos\theta - 1)(\tan\theta + 1) = 0$

e) $\sin^2\theta - 1 = 0$

f) $2\cos^2\theta - 1 = 0$

g) $\tan^2\theta - 3 = 0$

h) $4\sin^2\theta - 3 = 0$

8. Solve the quadratic trigonometric equations for θ, where $0° \leq \theta < 360°$.

a) $2\sin^2\theta + \sin\theta - 1 = 0$

b) $\cos^2\theta - \cos\theta - 2 = 0$

c) $\tan^2\theta - 2\tan\theta - 3 = 0$

d) $2\sin^2\theta - 3\sin\theta + 1 = 0$

e) $4\cos^2\theta - 1 = 0$

f) $2\sin^2\theta - \sin\theta = 0$

g) $\cos^2\theta + 2\cos\theta + 1 = 0$

h) $3\tan^2\theta - 1 = 0$

6 Reasoning and Analyzing Problems

Each statement is either **always true, sometimes true,** or **never true**. When you decide on your answer, you must provide a **justification**.

If you choose **sometimes true**, you must show one scenario when it is true, and another, when it is false.

For **always true** and **never true**, you must provide a brief explanation why. Sometimes this can simply be a definition, or sometimes it may require some more mathematical reasoning.

Example 6.1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

a) If SAS is given, then there is two possible triangles.

b) If $90° < \theta < 270°$, then $\cos\theta < 0$.

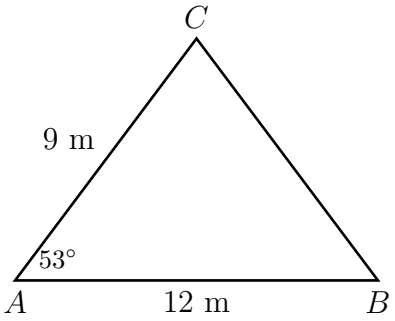
c) If $90° < \theta < 270°$, then $\tan\theta < 0$.

d) If $180° < \theta < 360°$, then $\sin\theta > 0$.

e) $\sin(-\theta) = -\sin\theta$

f) $\cos(-\theta) = \cos\theta$

Example 6.2. A student was using the Cosine Law to solve for side a. Identify and explain which step had an error. Make the necessary correction.



$a^2 = 9^2 + 12^2 - 2(9)(12)\cos 53°$	**Step 1**
$a^2 = 225 - 216\cos 53°$	**Step 2**
$a^2 = 9\cos 53°$	**Step 3**
$a = \sqrt{9\cos 53°}$	**Step 4**
$a \approx 2.33$	**Step 5**

Example 6.3. In each of the following, one item doesn't belong...

a) Which angle doesn't belong? Explain.

i. $45°$ ii. $135°$ iii. $225°$ iv. $330°$

b) Which expression doesn't belong? Explain.

i. $\cos 60°$ ii. $\sin 60°$ iii. $\cos 30°$ iv. $\sin 120°$

Practice Exercises

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) $\tan\theta = \tan(180° + \theta)$

   (b) If $\sin\theta = 0.5$, then there are two possible answers for θ.

   (c) If $0° \leq \theta < 360°$, then $\cos\theta > 1$.

   (d) If $0° \leq \theta < 360°$, then $\tan\theta > 1$.

   (e) If $90° < \theta < 180°$, then $\sin\theta > \cos\theta$.

   (f) Given $-1 < A < 1$ and $0° \leq \theta < 360°$, if $\cos\theta = A$, then there are two possible answers for θ.

2. A student was solving for θ. Identify and explain which step had an error. Make the necessary correction.

$$
\begin{aligned}
2\sin\theta + 1 &= 0 \quad \text{for all } \theta \\
2\sin\theta &= -1 && \textbf{Step 1} \\
\sin\theta &= -\frac{1}{2} && \textbf{Step 2} \\
\theta &= 240^\circ, 300^\circ && \textbf{Step 3} \\
\theta &= 240^\circ + 360n, 300^\circ + 360n, \quad n \in \mathbb{Z} && \textbf{Step 4}
\end{aligned}
$$

3. In each of the following, one item doesn't belong...

   a) Which angle doesn't belong? Explain.

   i. 390° ii. 150° iii. 330° iv. 30°

   b) Which expression doesn't belong? Explain.

   i. $\cos 0^\circ$ ii. $\sin 90^\circ$ iii. $\sin 270^\circ$ iv. $\tan 225^\circ$

7 Answer Key

Section 1 - Exercises

1. a) $Area \approx 15.09$ m^2 b) $Area \approx 31.33$ cm^2 c) $Area \approx 39.36$ ft^2
   d) $Area \approx 19.03$ in^2 e) $Area \approx 54.34$ ft^2 f) $Area \approx 15.59$ in^2

2. a) ≈ 8.50 in b) ≈ 22.64 cm c) ≈ 11.09 m d) ≈ 14.00 yd

3. a) $\theta \approx 54.3°$ b) $\theta \approx 75.7°$ c) $\theta \approx 43.3°$ d) $\theta \approx 82.9°$

4. $\angle B \approx 43.3°$
   $\angle C \approx 100.7°$
   $c \approx 10.03$ m

5. $\angle C \approx 36.2°$
   $\angle A \approx 43.8°$
   $a \approx 7.03$ m

Section 2 - Exercises

1. a) ≈ 6.06 cm b) ≈ 11.02 in c) ≈ 14.94 m d) ≈ 6.17 ft

2. a) $\theta \approx 85.5°$ b) $\theta \approx 75.5°$ c) $\theta = 90°$ d) $\theta \approx 80.4°$

3. $a \approx 7.09$ ft
   $\angle B \approx 102.6°$
   $\angle C \approx 38.4°$

4. The largest angle is approximately 100.0°.

5. The ships are about 38.42 nautical miles apart.

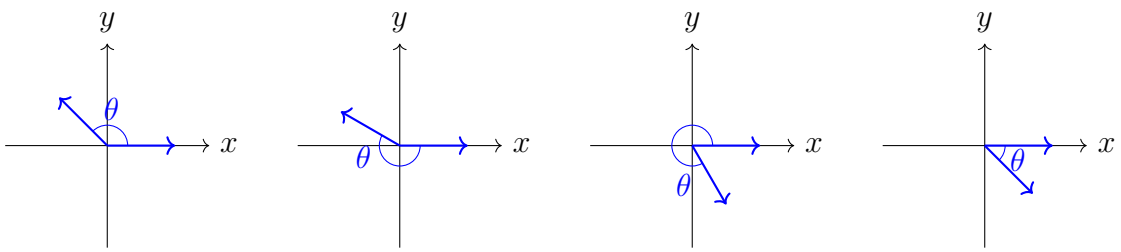
6. The pitcher's mouns is approximately 63.72 ft from first base.

7. The fence should make an angle of approximately 55.8° with the 120 m side.

8. The aircraft is 250 km from its starting point.

Section 3 - Exercises

1. a) b) c) d)



2. Answers may vary.

a) $405°, -315°$ b) $210°, -510°$ c) $-45°, 675°$ d) $285°, -435°$

3. a) $\theta_r = 30°$ b) $\theta_r = 60°$ c) $\theta_r = 45°$ d) $\theta_r = 30°$
e) $\theta_r = 30°$ f) $\theta_r = 30°$ g) $\theta_r = 80°$ h) $\theta_r = 40°$

4. a) $\theta = 30°, 150°, 210°, 330°$ b) $\theta = 55°, 125°, 235°, 305°$
c) $\theta = 15°, 165°, 195°, 345°$ d) $\theta = 72°, 108°, 252°, 288°$

5. a) $\theta_r = 40°$
$\theta = 140°, 220°, 320°$
b) $\theta_r = 30°$
$\theta = 150°, 210°, 330°$
c) $\theta_r = 20°$
$\theta = 160°, 200°, 340°$
d) $\theta_r = 60°$
$\theta = 120°, 240°, 300°$

6. a) $\sin\theta = -\frac{4}{5}$, $\cos\theta = \frac{3}{5}$, $\tan\theta = -\frac{4}{3}$
b) $\sin\theta = \frac{12}{13}$, $\cos\theta = -\frac{5}{13}$, $\tan\theta = -\frac{12}{5}$
c) $\sin\theta = -\frac{1}{\sqrt{2}}$, $\cos\theta = -\frac{1}{\sqrt{2}}$, $\tan\theta = 1$
d) $\sin\theta = \frac{1}{2}$, $\cos\theta = \frac{\sqrt{3}}{2}$, $\tan\theta = \frac{1}{\sqrt{3}}$

7. a) Quadrant 2 b) Quadrant 1 c) Quadrant 3
d) Quadrant 4 e) Quadrant 3 f) Quadrant 2

8. a) $\cos\theta = -\frac{4}{5}$, $\tan\theta = -\frac{3}{4}$
b) $\sin\theta = -\frac{5}{13}$, $\tan\theta = \frac{5}{12}$
c) $\sin\theta = -\frac{4}{5}$, $\cos\theta = -\frac{3}{5}$
d) $\cos\theta = -\frac{12}{13}$, $\tan\theta = \frac{5}{12}$
e) $\sin\theta = -\frac{7}{\sqrt{85}}$, $\cos\theta = \frac{6}{\sqrt{85}}$
f) $\sin\theta = -\frac{\sqrt{13}}{7}$, $\tan\theta = -\frac{\sqrt{13}}{6}$

Section 4 - Exercises

1. a) $\frac{\sqrt{3}}{2}$ b) $-\frac{\sqrt{3}}{2}$ c) $-\frac{\sqrt{3}}{2}$
   d) $-\frac{\sqrt{3}}{2}$ e) $-\frac{\sqrt{3}}{2}$ f) $\frac{\sqrt{3}}{2}$
   g) -1 h) 1 i) -1

2. a) $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ b) $-\frac{\sqrt{3}}{2}$ c) $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
   d) $-\frac{1}{2}$ e) $-\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$ f) $\sqrt{3}$
   g) $-\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$ h) $-\frac{1}{2}$ i) -1
   j) $\frac{1}{2}$ k) $-\frac{\sqrt{3}}{2}$ l) $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

3. a) 1 b) -1 c) *undefined* d) 0
   e) 0 f) 0 g) 0 h) 0
   i) 0 j) -1 k) 1 l) *undefined*

4. a) 2 triangles b) No triangle c) 2 triangles d) 1 triangle
   e) 1 triangle f) 2 triangles g) 1 triangle h) No triangle

5. a) Triangle 1: $\angle B \approx 53.5°, \angle C \approx 86.5°, c \approx 12.42$ cm
   Triangle 2: $\angle B \approx 126.5°, \angle C \approx 13.5°, c \approx 2.90$ cm
   b) Triangle 1: $\angle C \approx 45.8°, \angle A \approx 99.2°, a \approx 20.65$ cm
   Triangle 2: $\angle C \approx 134.2°, \angle A \approx 10.8°, a \approx 3.92$ cm
   c) Triangle 1: $\angle B \approx 42.5°, \angle C \approx 112.5°, c \approx 10.93$ cm
   Triangle 2: $\angle B \approx 137.5°, \angle C \approx 17.5°, c \approx 3.57$ cm
   d) Triangle 1: $\angle B \approx 66.8°, \angle C \approx 63.2°, a \approx 11.65$ cm
   Triangle 2: $\angle B \approx 113.2°, \angle C \approx 16.8°, a \approx 3.78$ cm

Section 5 - Exercises

1. a) $\theta = 60°, 120°$ b) $\theta = 60°, 300°$ c) $\theta = 45°, 225°$
   d) $\theta = 210°, 330°$ e) $\theta = 135°, 225°$ f) $\theta = 120°, 300°$
   g) $\theta = 45°, 135°$ h) $\theta = 120°, 240°$ i) $\theta = 60°, 240°$
   j) $\theta = 240°, 300°$ k) $\theta = 30°, 330°$ l) $\theta = 135°, 315°$

2. a) $\theta \approx 53.1°, 126.9°$ b) $\theta \approx 116.7°, 243.3°$ c) $\theta \approx 68.2°, 248.2°$
   d) $\theta \approx 199.3°, 340.7°$ e) $\theta \approx 23.1°, 336.9°$ f) $\theta \approx 143.1°, 323.1°$
   g) $\theta \approx 8.6°, 171.4°$ h) $\theta \approx 132.8°, 227.2°$ i) $\theta \approx 79.1°, 259.1°$

3. a) $\theta = 30° + 360n, 150° + 360n,$ $n \in \mathbb{Z}$
   b) $\theta = 150° + 360n, 210° + 360n,$ $n \in \mathbb{Z}$
   c) $\theta = 45° + 360n, 225° + 360n,$ $n \in \mathbb{Z}$
   d) $\theta = 225° + 360n, 315° + 360n,$ $n \in \mathbb{Z}$
   e) $\theta = 45° + 360n, 315° + 360n,$ $n \in \mathbb{Z}$
   f) $\theta = 120° + 360n, 300° + 360n,$ $n \in \mathbb{Z}$
   g) $\theta = 90° + 360n, 270° + 360n,$ *or* $\theta = 90° + 180n$ $n \in \mathbb{Z}$
   h) $\theta = 180° + 360n,$ $n \in \mathbb{Z}$

4. a) $\theta = -150°, -30°$
   b) $\theta = 30°, -30°$
   c) $\theta = 135°, -45°$
   d) $\theta = 60°$
   e) no solutions
   f) $\theta = 60°, -120°$

5. a) $\theta = 0°$ b) $\theta = 0°, 180°$ c) $\theta = 0°, 180°$
   d) $\theta = 90°$ e) $\theta = 180°$ f) $\theta = 90°, 270°$

6. a) $\theta = 30°, 150°$
   b) $\theta = 150°, 210°$
   c) $\theta = 45°, 135°$
   d) $\theta = 120°, 240°$
   e) $\theta = 270°$
   f) $\theta = 45°, 315°$
   g) $\theta = 120°, 300°$
   h) $\theta = 60°, 120°$

7. a) $\theta = 0°, 180°$
   b) $\theta = 0°, 180°, 270°$
   c) $\theta = 30°, 90°, 150°, 270°$
   d) $\theta = 60°, 135°, 300°, 315°$
   e) $\theta = 90°, 270°$
   f) $\theta = 45°, 135°, 225°, 315°$
   g) $\theta = 60°, 120°, 240°, 300°$
   h) $\theta = 60°, 120°, 240°, 300°$

8. a) $\theta = 30°, 120°, 240°, 300°$
   b) $\theta = 180°$
   c) $\theta = 135°, 225°, \theta \approx 71.6°, 251.6°$
   d) $\theta = 30°, 90°, 150°$
   e) $\theta = 60°, 120°, 240°, 300°$
   f) $\theta = 30°, 90°, 150°, 180°$
   g) $\theta = 180°$
   h) $\theta = 30°, 150°, 210°, 330°$

Section 6 - Exercises

1. a) Always true. If θ_r is in Q1, then $180 + \theta_r$ would be in Q3. Since tangent is both positive in Q1 and Q3, or negative in Q2 and Q4. Note that Q2 is usually $180 - \theta$, and Q4 is $360 - \theta$. So the difference from Q4 to Q2, is also $180°$.

   b) Sometimes true. It depends on what domain of θ we are considering. If there are no restrictions, then there are infinite solutions. If there is between $[0, 360°]$, then there are two.

   c) Never true. Cosine's ratio is $\frac{x}{r}$, and x is never greater than r, so it's not possible.

   d) Sometimes true. Since tangent's ratio is $\frac{y}{x}$, there are points on the circle where $y > x$, and points on the circle where $y \leq x$.

   e) Always true. The angles are in quadrant 2, and in that quadrant, cosine is always negative, whereas sine is always positive.

   f) Always true. Cosine always has two answers between -1 and 1. Only when $A = \pm 1$, does cosine only have one solution in that domain.

2. Step 3 was incorrect. The correct reference angle should be $30°$, so the corresponding quadrant angles are $210°$ and $330°$.

3. a) i. doesn't belong because it's not a primary angle.

   b) iii. doesn't belong because it's the only one that is -1
   iv. doesn't belong because it's the only one with a non-quadrantal angle.